

# SOFTWARE ENGINEERING

## LAB 4

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**Chat link -<https://chatgpt.com/share/6ac4faa2-d468-83ee-abd2-7639038f4f45>**

### **Task 1 — Fix the arm movement**

This is a Pygame arm wrestling game. Alternating Left/Right arrow keys currently pushes the arm toward the computer's side instead of pulling it toward the player.

The win condition is:

- -100 means the player wins
- +100 means the computer wins

Find the bug and give me the minimal fix, showing only the changed lines and where they go.

### **Task 2 — AI stamina cycle**

Now add an AI stamina cycle. The computer should periodically enter a brief high-power surge with increased pushing force, followed by a cooldown with reduced force.

Use a timer-based state machine with states:

- NORMAL
- SURGE
- COOLDOWN

Put the durations and force multipliers in named constants.

Give me the complete updated code for the changed methods only.

### **Task 3 — Visual warnings**

Building on the previous task, add visual cues:

1. A flashing "AI SURGE!" warning while the AI is in its surge state.
2. An "EXHAUSTED!" indicator whenever the player's stamina is below 10 and input is halted.

Keep all drawing code in the existing draw/render method, and don't change the game logic.

### **Task 4 — Counter-surge**

Building on the previous tasks: if the player pushes within a short window right after the AI's surge ends and its cooldown begins, grant a temporary stamina recovery boost and double the player's push strength for a few seconds.

Show a "COUNTER-SURGE!" indicator while the bonus is active.

Use named constants for the window and bonus duration.